

Chandrayaan-4: South Polar Landing Site - Detailed Summary

1. Mission Overview

- Chandrayaan-4 is India's first lunar sample-return mission.
- The mission aims to collect Moon soil and rock samples and bring them back to Earth.

2. Landing Site Selection

- ISRO selected a site in the Moon's south polar region.
- The specific region chosen is Mons Mouton.
- Four candidate sites were studied: MM-1, MM-3, MM-4, MM-5.
- MM-4 was selected as the safest and most suitable site.

3. Terrain & Safety Criteria

- High-resolution data from Orbiter High Resolution Camera (OHRC) was used.
- The selected site has low hazard percentage.
- The terrain has gentle slopes (~5 degrees).
- Multiple 24m x 24m hazard-free landing grids were identified.

4. Mission Modules

- Propulsion Module (PM) – Provides thrust.
- Descender Module (DM) – Lands on the Moon.
- Ascender Module (AM) – Lifts off with collected samples.
- Transfer Module (TM) – Transfers samples to Earth orbit.
- Re-entry Module (RM) – Brings samples safely back to Earth.

5. Importance of the Mission

- Enhances India's space exploration capability.
- Enables detailed scientific study of lunar materials.
- Supports future lunar missions and exploration plans.